FIITJEE REVIEW MOCK TEST-02

(BATCHES: 921 LOT)

IIT – JEE 2021

Time: 1 Hours **Maximum Marks: 66**

A. Question Paper Format

1. The question paper consists three parts.
2. **Part A – (I)** contains **6** multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part A – (II)** contains **6** multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are correct**.
4. **Part-B (01-05)** contains five (05) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

B. Marking scheme:

5. For each question in Part A (I), you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.
6. For each question in Part A (II), you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.
   Partial Marks: **+1** For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.
   Zero Marks: **0** If none of the bubbles is darkened.
   ***Negative Marks*: −2 In all other cases.**
   For example, if **(A), (C) and (D)** are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only **(A) and (D)** will result in **+2 marks**; and darkening **(A) and (B)** will result in **−2** marks, as a wrong option is also darkened.
7. For each question in Part B, you will be awarded **+4 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | | | | | |

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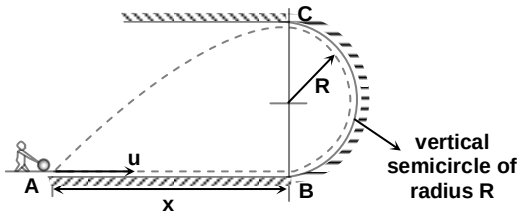
PHYSICS

PART – A(I) (Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. Velocity of a body of mass 2kg moving in x-y plane is given by $\vec{v} = \left(2\hat{i} + 4t\hat{j}\right)$ m/s, where t is the time in second. The power delivered to the body by the resultant force acting on it at t = 5 sec is.
   (A) 80 W (B) 160 W
   (C) 40 W (D) 100 W

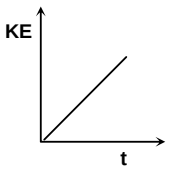
2. A person rolls a small ball with speed u along the floor from point A. If x = 3R, determine the required speed u so that the ball returns to A after rolling on the circular surface in the vertical plane from B to C and becoming a projectile at C. (Neglect friction and radius of ball)



   (A) $\frac{3}{2}\sqrt{gR}$ (B) $\frac{1}{2}\sqrt{gR}$
   (C) $\frac{5}{2}\sqrt{gR}$ (D) $\frac{7}{2}\sqrt{gR}$

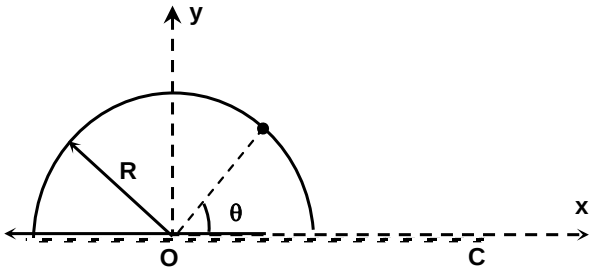
3. Two astronauts, each of mass 75 kg are floating next to each other in space, outside the space shuttle one of them pushes the other through a distance of 1 m (relative to himself) with a force of 300 N. What is the final relative velocity of the two?
   (A) 2.0 m/s (B) 2.83 m/s
   (C) 5.66 m/s (D) 4 m/s

4. The kinetic energy of a body moving along a straight line varies with time as shown in the figure. The force acting on the body is
   (A) zero
   (B) constant
   (C) directly proportional to velocity
   (D) inversely proportional to velocity.



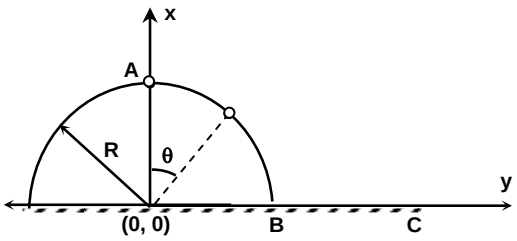
Space for rough work

5. A thin semicircular disc of non uniform mass distribution is shown in x-y plane. Surface mass density of disc varies according to relation $\sigma = \sigma_o \sin\theta$, where σ_o is a positive constant and θ is the angle made by position vector of any point on disc with positive x-axis. The coordinates of centre of mass of semicircular disc are



(A) $(0, 2R/\pi)$
(B) $(0, \pi R/6)$
(C) $(0, 3\pi R/2)$
(D) $(\pi R/2, 3\pi R/4)$

6. A small particle of mass m is placed at top point A of fixed smooth hemisphere of radius R. Particle is given small displacement towards right and it starts slipping. Collision of particle with frictionless horizontal surface is perfectly inelastic. Velocity of particle after hitting horizontal surface will be



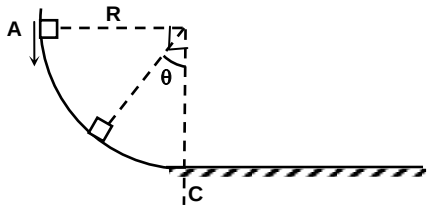
(A) $\sqrt{gR}$
(B) $\sqrt{\frac{2}{3}gR}$
(C) $\frac{2}{3}\sqrt{\frac{2}{3}gR}$
(D) $\sqrt{\frac{gR}{3}}$

Space for rough work

PART – A(II) (Multiple Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

7. A particle of mass m is placed on vertical fixed circular track and then it is given velocity v vertically downward at position A on track. If block moves on track with constant speed then



(A) Coefficient of friction between block and circular track as function of angle θ is $\mu = \dfrac{\sin\theta}{\cot\theta + \dfrac{v^2}{Rg}}$

(B) Coefficient of friction between block and circular track as function of angular θ is $\mu = \dfrac{\sin\theta}{\cos\theta + \dfrac{v^2}{Rg}}$

(C) Instantaneous power due to friction is ($-mg \sin\theta\, v$)

(D) Work done from A to C by friction on block will be ($-mgR$)

8. The potential energy of a particle of mass m at a distance r from a fixed point O is given by $V(r) = kr^2/2$, where k is a positive constant of appropriate dimensions. This particle is moving in a circular orbit of radius R about the point O. If v is the speed of the particle and L is the magnitude of its angular momentum about O, which of the following statements is (are) true?

(A) $v = \sqrt{\dfrac{k}{2m}}R$

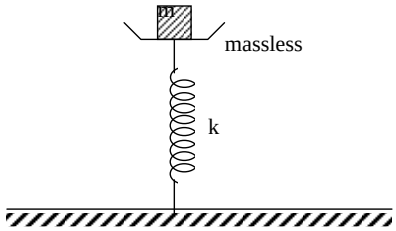
(B) $v = \sqrt{\dfrac{k}{m}}R$

(C) $L = \sqrt{mk}R^2$

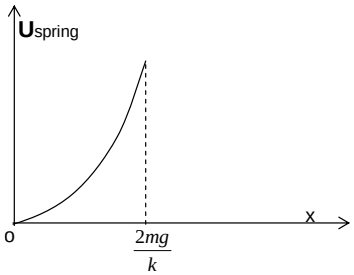
(D) $L = \sqrt{\dfrac{mk}{2}}R^2$

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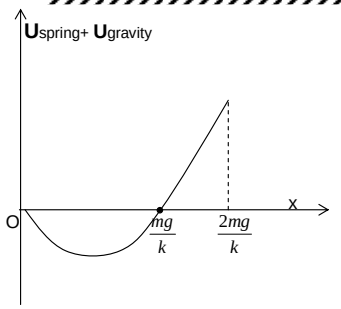
9. A block of mass m is placed on a massless spring (initially unstretched) and released. Due to mass, oscillation starts. There is no loss of energy and initially gravitational potential energy ($U_{gravity}$) is assume to be 0. Potential energy U will change with displacement x of the block as



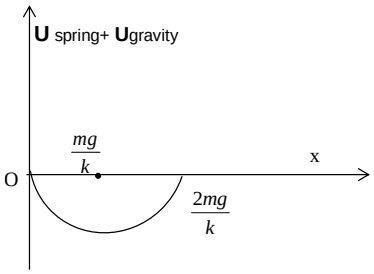
(A)



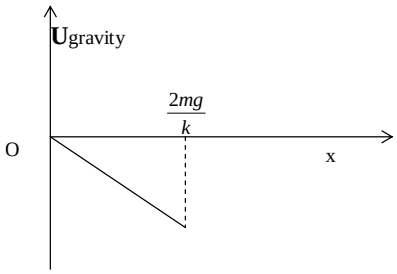
(B)



(C)



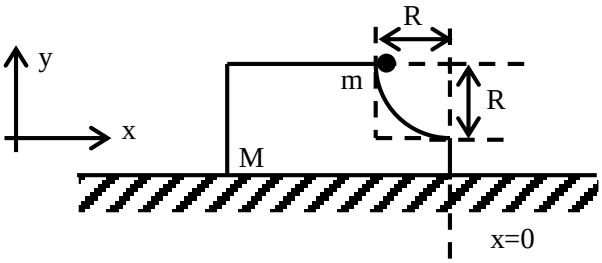
(D)



10. A ball moving with kinetic energy 3J makes an elastic head-on collision with a stationary ball which has twice its mass. During the impact,
(A) the minimum kinetic energy of the system is 1.5 J
(B) the maximum elastic potential energy of the system is 1.5 J
(C) momentum and total energy are not conserved at every instant
(D) the ratio of kinetic energy to potential energy of the system first decreases and then increases

Space for rough work

11. A block of mass M has a circular cut with a frictionless surface as shown. The block rests on the horizontal frictionless surface of a fixed table. Initially the right edge of the block is at x = 0, in a co-ordinate system fixed to the table. A point mass m is released from rest at the topmost point of the path as shown and it slides down. When the mass loses contact with the block, its position is x and the velocity is v. At that instant, which of the following options is/are correct?



(A) The x component of displacement of the center of mass of the block M is : $-\frac{mR}{M+m}$.

(B) The position of the point mass is : $x = -\sqrt{2}\frac{mR}{M+m}$.

(C) The velocity of the point mass m is : $v = \sqrt{\frac{2gR}{1+\frac{m}{M}}}$.

(D) The velocity of the block M is: $V = -\frac{m}{M}\sqrt{2gR}$.

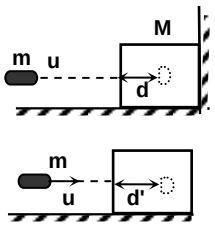
12. A point mass of 1 kg collides elastically with a stationary point mass of 5 kg. After their collision, the 1 kg mass reverses its direction and moves with a speed of 2 ms^{-1}. Which of the following statement(s) is (are) correct for the system of these two masses?
(A) Total momentum of the system is 3 kg ms^{-1}
(B) Momentum of 5 kg mass after collision is 4 kg ms^{-1}
(C) Kinetic energy of the centre of mass is 0.75 J
(D) Total kinetic energy of the system is 4J

Space for rough work

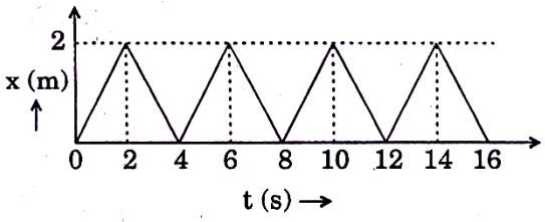
SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

13. A bullet of mass m fixed with velocity u penetrates upto thickness 'd' into the block, if block is fixed. Now if block is free to move on smooth ground and bullet is fired into it symmetrically with same velocity u it penetrates upto thickness d'. Assume resistance force due to block on bullet in both case is same. Find ratio $\frac{d}{d'}$ if $m = \frac{M}{4}$.

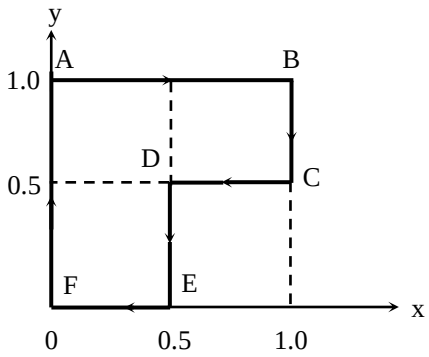


14. The figure shows the position – time (x – t) graph of one-dimensional motion of a body of mass 0.225 kg. The magnitude of each impulse (in Ns) is

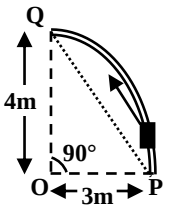


15. A boy of mass 50 kg is standing on a 150 kg free to move long cart. There is negligible friction between cart and ground. Initially, the boy is standing 5 m from a wall. If he walks 3 m on the cart towards the wall, then the final distance(in m) of the boy from the wall will be

16. A particle is moved along a path AB-BC-CD-DE-EF-FA, as shown in figure, in presence of a force $\vec{F} = \left(\alpha y\hat{i} + 2\alpha x\hat{j}\right)N$, where x and y are in meter and α = – 1 Nm^{-1}. The work done on the particle by this force $\vec{F}$ will be _____ Joule.



17. Consider an elliptically shaped rail PQ in the vertical plane with OP = 3 m and OQ = 4 m. A block of mass 0.125 kg initially at rest is pulled along the rail from P to Q with a force of 2.25 N, which is always parallel to line PQ (see the figure given). Assuming no frictional losses, the kinetic energy (in joule) of the block when it reaches Q is (take acceleration due to gravity = 10 ms^{-2})



18. A block of mass 0.50 kg is moving with a speed of 2.00 m/s on a smooth surface. It strikes another mass of 1.5 kg and then they move together as a single body. The energy loss(in joule) during the collision is

Space for rough work

FIITJEE REVIEW MOCK TEST-02

(BATCHES: 921 LOT)

IIT – JEE 2021

ANSWERS

PHYSICS

1.	**B**	2.	**C**	3.	**D**	4.	**D**
5.	**B**	6.	**C**	7.	**B, C, D**	8.	**B, C**
9.	**A, C, D**	10.	**D**	11.	**A, C**	12.	**A, C**
13.	**1.25**	14.	**0.45**	15.	**2.75**	16.	**0.75**
17.	**6.25**	18.	**0.75**				

SOLUTIONS

1. Power = $\frac{d(K.E.)}{dt}$

2. We determine the velocity at C so that it reaches A along a parabolic trajectory; then applying the law of conservation of energy to the motion ABC (as non-conservative force are absent)

   v = velocity at point C = $\frac{3}{2}\sqrt{gR}$

   applying conservation of energy from A to C

   $\frac{1}{2}mu^2 = 2mgR + \frac{1}{2}mv^2$

   $\Rightarrow u = \frac{5}{2}\sqrt{gR}$

3. Work done by one astronaut on another astronaut is $300 \times 0.5 = \frac{1}{2}mv^2$

   v = 2 m/s
   relative velocity = 4 m/s

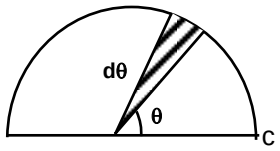
4. $\frac{1}{2}mv^2 = pt$ (P = const)

   $\Rightarrow$ $v = \sqrt{\frac{2Pt}{m}}$

   $a = \frac{dv}{dt} = \sqrt{\frac{2P}{m}}\frac{1}{2\sqrt{t}}$

   $F = ma = \sqrt{\frac{mP}{2t}} \propto \frac{1}{v}$

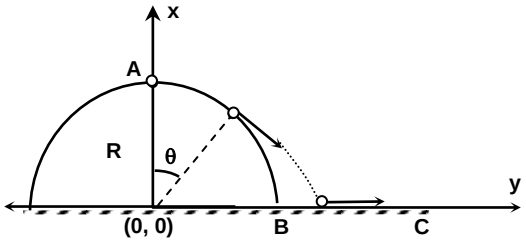
5. Take sector element on disc



$$Y_{cm} = \frac{\int_0^{\pi}(\sigma_o \sin\theta)\frac{R^2 d\theta}{2} \times \left(\frac{2R}{3}\sin\theta\right)}{\int_0^{\pi}(\sigma_o \sin\theta)\frac{R^2 d\theta}{2}} = \frac{\frac{2}{3}\int_0^{\pi} R\sin^2\theta d\theta}{\int_0^{\pi}\sin\theta d\theta} = \frac{\pi R}{6}$$

   $X_{cm} = 0$

6. At the line of leaving sphere horizontal component of velocity is $\frac{2}{3}\sqrt{\frac{2}{3}gR}$ and it will remain constant.



7. $f = mg\sin\theta$
   $N - mg\cos\theta = mv^2/R$

   $f = \mu N \Rightarrow \mu = \frac{\sin\theta}{\cos\theta + \frac{v^2}{Rg}}$

8. $V = \frac{kr^2}{2} \Rightarrow F = -\frac{dV}{dr} = -kr$

$\therefore \frac{mv^2}{r} = kr \Rightarrow v = \sqrt{\frac{k}{m}}R$

Angular momentum L = mvr = $m\sqrt{\frac{k}{m}}R^2 = \sqrt{mk}R^2$

10. In a head-on elastic collision between two particles, the kinetic energy becomes minimum and the potential energy becomes maximum at the instant when the two move with a common velocity. The momentum and the total energy are conserved at every instant.
Let m and u be the mass and initial velocity respectively of the first particle, 2m be the mass of the second particle, and v be the common velocity. Then

$\frac{1}{2}mu^2 = 3$ J.

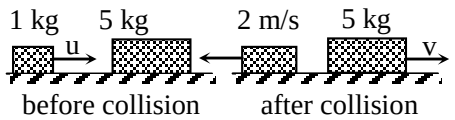
$mu = (m + 2m)v$ or $v = \frac{u}{3}$.

$\therefore$ the minimum KE of the system = $\frac{1}{2}(3m)\left(\frac{u}{3}\right)^2 = \left(\frac{1}{2}mu^2\right)\frac{1}{3} = 1$J.

$\therefore$ the maximum PE of the system = 2J.

11. Δx_{cm} of the block & point mass system = 0
$\therefore$ m (x + R) + Mx = 0
where x is displacement of the block.
$x = -\frac{mR}{M+m}$
From conservation of momentum and mechanical energy of the combined system
0 = mv – MV
$mgR = \frac{1}{2}mv^2 + \frac{1}{2}MV^2$
$\therefore v = \sqrt{\frac{2gR}{1+\frac{m}{M}}}$.

12. By conservation of linear momentum
$u = 5v - 2$ …(i)
By Newton's experimental law of collision
$u = v + 2$ …(ii)
using (i) and (ii) we have
v = 1 m/s and u = 3 m/s
Kinetic energy of the centre of mass =
$\frac{1}{2}m_{system}v_{cm}^2 = 0.75$ J



16. $w_{AB} = \int_0^1 \alpha y dx = -1$

$w_{BC} = \int_1^{0.5} 2\alpha x dy = +1$

$w_{CD} = \int_1^{0.5} \alpha y dx = +0.25$

$w_{DE} = \int_{0.5}^{0} 2\alpha x dy = +0.5$

$w_{EF} = w_{FA} = 0$

$W_{net} = 0.75$

FIITJEE REVIEW MOCK TEST-02

CHEMISTRY

(Batches: 921 Lot)

IIT-JEE, 2021

Time: 1 Hour **Maximum Marks: 66**

INSTRUCTIONS

A. Question Paper Format

1. The question paper consists of four parts.
2. **Part-A – (01-06)** contains **06 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part-A – (07-12)** contains **06 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are correct**.
4. **Part-C – (13-18)** contains **06 questions**. The answer to each question is a **Numerical Decimal based**, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

B. Marking scheme:

(i). For each question in **Part A (2)**, you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.

(ii). For each question in **Part A (3),** you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.

Partial Marks : + 1 For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks: – 2 In all other cases.

For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get –2 marks, as a wrong option is also darkened.

(iii). For each question in **Part C (4),** you will be awarded **+4 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | |

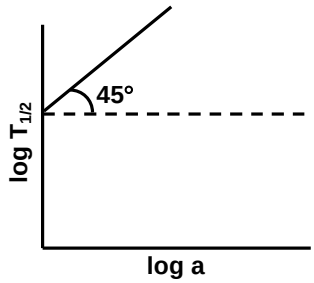
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PART – A(I): (Single Correct Answer Type)

*This section contains **06 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.***

1. Following is the graph between log T_{50} & log a (a = initial concentration) for a given reaction at 27°C. Hence order is



(A) 0 (B) 1
(C) 2 (D) 3

2. Consider the following reactions at 300 K.

$A \longrightarrow B$ (uncatalysed reaction)

$A \xrightarrow{\text{catalyst}} B$ (catalyst reaction)

The activation energy is lowered by 8.314 KJ mol^{-1} for the catalysed reaction. How many times the rate of this catalysed reaction greater than that of uncatalysed reaction? (Given $e^{3.33} = 28$)

(A) 15 times (B) 38 times
(C) 22 times (D) 28 times

3. Identify the statement which is **correct** from the following.
   (A) Rate of reaction increases substantially with increase in temperature since frequency of collision increases.
   (B) Rate of reaction decreases on increasing temperature if the reaction is exothermic.
   (C) Rate of reaction remains unchanged if temperature changes for reactions having no activation energy.
   (D) Rate of reactions decreases on decreasing temperature only if the reaction is endothermic.

4. In the sequence of the following

$_{92}Xe^{238} \xrightarrow{-\alpha} Y \xrightarrow{-\beta} Z \xrightarrow{-\beta} L \xrightarrow{-n\alpha} {}_{84}M^{218}$ the value of n will be :

(A) 5 (B) 4
(C) 3 (D) 6

Space for rough work

5. The reaction $A(g) \longrightarrow B(g) + 2C(g)$ is a first order reaction with rate constant 2.772×10^{-3} s^{-1}. Starting with 0.1 mole of A in 2 litre vessel, find the concentration of A after 250 sec when the reaction is allowed to take place at constant pressure and at 300 K?
(A) 0.0125 M (B) 0.025 M
(C) 0.05 M (D) none of these

6. Isodiaphers are atoms having
(A) *n*/*p* same (B) *p*/*n* same
(C) (*n* –*p*) same (D) (*n*-*p*) different

PART – A(II): (Multi Correct Answer Type)

*This section contains **06 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.*

7. For a complex reaction:
(A) order of overall reaction is same as the molecularity of slowest step.
(B) order of overall reaction is less than the molecularity of slowest step.
(C) order of overall reaction is greater than the molecularity of slowest step.
(D) molecularity of slowest step is neither zero nor fractional.

8. Select the correct statement(s) for position emission by unstable nucleus.
(A) X-ray emission takes place (B) A neutron is formed
(C) $\frac{n}{p}$ of daughter nucleus increases (D) A neutron is consumed

9. The reaction rate of the reaction ; $H_2(g) + Br_2(g) \longrightarrow 2HBr(g)$; is given by Rate = $K[H_2]\,[Br_2]^{1/2}$
Which of the following statements is/are true?
(A) Order of reaction is 2 (B) Molecularly of reaction is 2
(C) Order of reaction is 1.5 (D) Molecularly of reaction is 1.5

10. If the rate of reaction between A and B is given by rate = $K[A][B]^2$, then the reaction is
(A) first order in A (B) second order in B
(C) third order overall (D) none of the above

Space for rough work

11. Which of the following statements are in accordance with the Arrhenius equation?
(A) Rate of a reaction increases with increase in temperature.
(B) Rate of a reaction increases with decrease in activation energy.
(C) Rate constant decreases exponentially with increase in temperature
(D) Rate of reaction decreases with decrease in activation energy.

12. Decrease in atomic number is observed during
(A) Alpha emission (B) Beta emission
(C) Positron emission (D) electron capture

PART-C (Numerical Decimal Based Type)

This section contains **6 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

13. A first order reaction is 50% completed in 30 minutes at 27° and in 10 minutes at 47°C. The energy of activation of the reaction is (Ans.kJ/mol)

14. Number of natural life times (T_{av}) required for first order reaction to achieve 99.9% level of completion is

15. If the half life of a radioactive sample is 2n years. Then the fraction of a sample which will remain decayed after n years is $\frac{1}{\sqrt{x}}$. Then the value of x will be

16. If the reaction A + B $\longrightarrow$ P is exothermic to the extent of 30 kcal/mole and E_a for the forward reaction is 294 kJ/mole, E_a for the backward reaction in kcal/mole is:

17. Consider the following first order competing reactions $X \xrightarrow{k_1} A + B$ and $Y \xrightarrow{k_2} C + D$. If 50% of the reaction of X was completed when 96% of the reaction of Y completed, the ratio of their rate constants (k_2 /k_1) is:

18. For a given reaction, energy of activation for forward reaction (E_{af}) is 80 kJ mol^{-1}. ΔH = – 40 kJ mol^{-1} for the reaction. A catalyst lowers E_{af} to 20 kJ mol^{-1}. The ratio of energy of activation for reverse reaction before and after addition of catalyst is:

Space for rough work

FIITJEE REVIEW MOCK TEST-02

CHEMISTRY

(Batches: 921 Lot)

IIT-JEE, 2021

ANSWER KEYS

1. **A**
2. **D**
3. **C**
4. **B**
5. **A**
6. **C**
7. **A, D**
8. **B, C**
9. **B, C**
10. **A, B, C**
11. **A, B**
12. **A, C, D**
13. **43.84**
14. **6.9**
15. **2.00**
16. **324**
17. **4.65**
18. **1.2**

FIITJEE REVIEW TEST-02

(Batches: 921 Lot)

IIT – JEE, 2021

MATHEMATICS

Time: 1 Hour **Maximum Marks: 66**

INSTRUCTIONS

A. Question Paper Format

1. The question paper consists of four parts.
2. **Part-A – (01-06)** contains **06 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part-A – (07-12)** contains **06 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are correct**.
4. **Part-C – (13-18)** contains **06 questions Numerical based questions**, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

B. Marking scheme:

(i). For each question in **Part A (2)**, you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.

(ii). For each question in **Part A (3),** you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.

Partial Marks : + 1 For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks: – 2 In all other cases.

For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get –2 marks, as a wrong option is also darkened.

(iii). For each question in **Part C (4),** you will be awarded **+4 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | |

Name : . . . . . .

Batch : . . . . . **Session :**. . . . . **Date:**. . . . . .

SECTION – III : MATHEMATICS

PART – A: (Single Correct Answer Type)

*This section contains **06 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.***

1. P lies on the line 8y – 15x = 0 and Q lies on the line 10y – 3x = 0 and the mid-point of PQ is (8, 6). The distance PQ is

   (A) $\frac{50}{7}$ (B) $\frac{60}{7}$

   (C) $\frac{50}{3}$ (D) $\frac{60}{3}$

2. The circles which can be drawn to pass through (3, 0) and (5, 0) and to touch the y-axis, intersect at an angle θ, then $\cos\theta$ is equal to

   (A) $\frac{7}{8}$ (B) $\frac{1}{2}$

   (C) 0 (D) none of these

3. A variable circle passes through the fixed point P(h, k) and touches x–axis. The locus of the other end of the diameter through P is

   (A) $(y-h)^2 = 4kx$ (B) $(x-k)^2 = 4hy$

   (C) $(x-h)^2 = 4ky$ (D) $(y-k)^2 = 4hx$

4. The locus of the centre of a circle which touches a given line and passes through a given point, not lying on the given line, is

   (A) a parabola (B) a circle

   (C) a pair of straight line (D) none of these .

5. The coordinates of the point on the parabola $y = x^2 + 7x + 2$, which is nearest to the straight line $y = 3x - 3$ are

   (A) (-2, -8) (B) (1, 10)

   (C) (2, 20) (D) (-1, -4)

Space for rough work

6. The circle drawn with variable chord $x + ay - 5 = 0$ (a being a parameter) of the parabola $y^2 = 20x$ as diameter will always touch the line
(A) $x + 5 = 0$ (B) $y + 5 = 0$
(C) $x + y + 5 = 0$ (D) $x - y + 5 = 0$

PART – A: (Multi Correct Answer Type)

*This section contains **06 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.*

7. ABCD is a rectangle $E(4,3), F(2,5)$ lie on AB and CD respectively such that EF divides the rectangle in two equal parts. If the co-ordinates of A is (7, 3), other co-ordinates of the vertices can be
(A) (7, 2) (B) (7, 5)
(C) (–1, 3) (D) (–1, 7)

8. If $\frac{x}{a}+\frac{y}{b}=1$ and $\frac{x}{c}+\frac{y}{d}=1$ intersect the axes at four concyclic points and $c^2 + a^2 = b^2 + d^2$, then these lines can intersect at (a, b, c, d > 0)
(A) (1, 1) (B) (1, – 1)
(C) (2, – 2) (D) (3, – 2)

9. The tangents drawn from the origin to the circle $x^2 + y^2 + 2gx + 2fy + f^2 = 0$ are perpendicular if
(A) g = f (B) g = -f
(C) g = 2f (D) 2g = f

10. The equation(s) of the tangent at the point (0, 0) to the circle, making intercepts of length 2a and 2b units on the coordinate axes, is (are)
(A) ax + by = 0 (B) ax – by = 0
(C) x = y (D) None of these

11. A quadrilateral is inscribed in a parabola, then
(A) quadrilateral may be cyclic .
(B) diagonals of the quadrilateral may be equal .
(C) all possible pairs of adjacent sides may be perpendicular .
(D) none of these.

12. The directrix of a parabola is $2x - y = 1$. The focus lies on y-axis. The locus of one end of the latus rectum is
(A) $7x - y - 1 = 0$ (B) $2x - y - 1 = 0$
(C) $2x + y + 1 = 0$ (D) $3x + y + 1 = 0$

Space for rough work

PART – C: (Numerical Based Type)

*This section contains **06 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).*

13. P is a point on the circumcircle of a triangle ABC. Foot of perpendicular from P on AB and AC are (0, 0) and (3, 3) respectively. If equation of BC is $3x + 2y + 5 = 0$ and the foot of perpendicular from P on BC is (a, b), then $|a + b|$ is equal to _____

14. Let $2x + 3y = 0$ and $3x - 2y = 0$ intersect at A. Let another line whose equation is $ax + by + c = 0$ intersects the given two lines at B and C respectively. A circle touches AB at A and BC at D. It intersects AC in E. If CE = 3, CD = 6, then BD = _____

15. Let $x, y \in R$ satisfy $x^2 + y^2 + 2x + 2y - 2 = 0$, $a = \max\{x^2 + y^2\}$ and $b = \min\{x^2 + y^2\}$. The average of a and b is equal to _____

16. For a given circle of radius 2, if s is the semi-perimeter of the triangle which inscribe maximum area on the circle, then [s] is _____

17. From origin tangents OA and OB are drawn to $(x - 2)^2 + (y - 2)^2 = 1$. If line PQ touches $(y + 3)^2 = 4a(x + 4)$, then $\frac{a}{7}$ is _____ (where P and Q are mid point of OA and OB)

18. Two triangles ABC and DEF are formed by three tangents and the corresponding three normals to a parabola $y^2 = 4ax$ respectively. The orthocenter of ΔABC has coordinates (–5, 4). Then, distance of orthocenter of ΔDEF from axis of parabola is _____

Space for rough work

FIITJEE REVIEW TEST

MATHEMATICS

ANSWERS

MATHEMATICS (SECTION –III)

PART – A

1. **B**	2. **A**	3. **C**	4. **A**
5. **A**	6. **A**	7. **B, C**	8. **A, B, C**
9. **A, B**	10. **A, B**	11. **A, B**	12. **A, D**

PART – C

13. **2.00**	14. **9.00**	15. **6.00**	16. **5.00**
17. **1.25**	18. **4.00**		

HINTS & SOLUTION

MATHEMATICS (SECTION –III)

PART – A

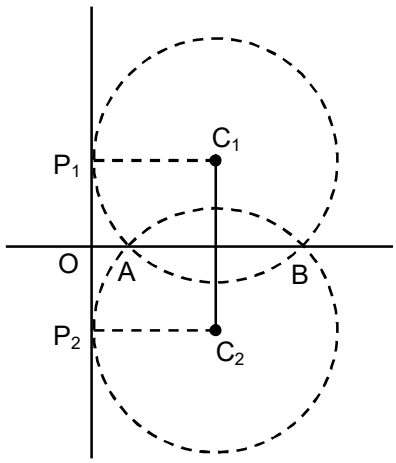
1. **B**

The line through (8, 6) and parallel to OQ is 10y = 3x + 36 and thus meets OP at $\left(\frac{16}{7}, \frac{30}{7}\right)$.

But this is mid point of OP. So P is $\left(\frac{32}{7}, \frac{60}{7}\right)$.

2. **A**

Let A ≡ (3, 0), B ≡ (5, 0) and C_1, C_2 be the centres of circles passing through A, B and touching the y-axis at P_1 and P_2.
Radius of both the circles will be same.
Let $C_1A = C_2A = r$
Now $C_1 \equiv \left(4, \sqrt{15}\right)$, $C_2 \equiv \left(4, -\sqrt{15}\right)$
If $C_1AC_2 = \alpha \Rightarrow \cos\alpha = -\frac{7}{8} \Rightarrow \cos\theta = \frac{7}{8}$.



3. **C**

Distance from (h, k) = distance from x–axis $(h - x)^2 + (k - y)^2 = (y + k)^2 \Rightarrow (x - h)^2 = 4ky$

4. **A**

The centre will be equidistant from the given line and the given point .

5. **A**

Any point on the parabola is $(x , x^2 + 7x + 2)$
Its distance from the line y = 3x –3 is given by

$$P = \left|\frac{3x - \left(x^2 + 7x + 2\right) - 3}{\sqrt{9+1}}\right| = \left|\frac{x^2 + 4x + 5}{\sqrt{10}}\right| = \frac{x^2 + 4x + 5}{\sqrt{10}}$$ (as $x^2 + 4x + 5 > 0$ for all $x \in R$)

$$\frac{dP}{dx} = 0 \Rightarrow x = -2$$

The required point ≡ (-2, -8).

6. **A**

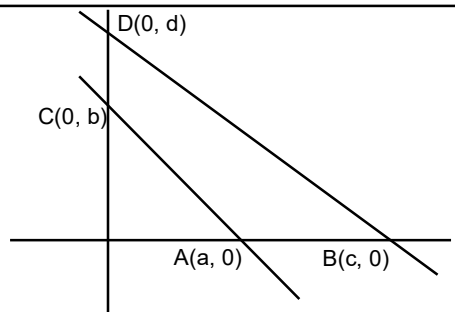
Clearly x + ay – 5 = 0 will always pass through the focus of $y^2 = 20x$ i.e. (5, 0).
Thus the drawn circle will always touch the directrix of the parabola i.e. the line x + 5=0

7. **B, C**

Mid point of EF is the centre of rectangle.

8. **A, B, C**

Points A, B, C, D are concyclic, then ac = bd.
The co-ordinate of the points of intersection of lines are $\left(\frac{ac(b-d)}{bc-ad}, \frac{bd(c-a)}{bc-ad}\right)$.

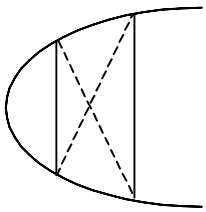


Let co-ordinates of the points of intersection is (h, k)
$h = \frac{ac(b-d)}{bc-ad}$, $k = \frac{bd(c-a)}{bc-ad}$
given $c^2 + a^2 = b^2 + d^2$ ($\because$ ac = bd)
$\Rightarrow (c-a)^2 = (b-d)^2$
$\Rightarrow c - a = \pm (b - d)$
then the locus of the points of intersection is $y = \pm x$.

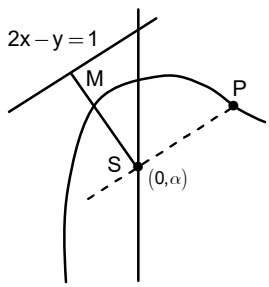
9. **A, B**
Radius of the given circle = |g| . For perpendicular tangents origin should lie on the director circle, i.e. distance of origin from the centre of the given circle = $|g|\sqrt{2}$
$\Rightarrow g^2 + f^2 = 2g^2 \Rightarrow g = \pm f$.

10. **A, B**
Equation of circle passing through origin and cutting off intercepts 2a and 2b units on the coordinate axes is
$x^2 + y^2 \pm 2ax \pm 2by = 0$
and equation of tangent at (0, 0) is
$ax \pm by = 0$

11. **A, B**
As a circle can intersect a parabola in four points, so quadrilateral may be cyclic. The diagonals of the quadrilateral may be equal as the quadrilateral may be an isosceles trapezium.
A rectangle cannot be inscribed in a parabola.
So (C) is wrong.



12. **A, D**
Let the focus be $(0, \alpha)$



$SM = SP \qquad SM = \frac{\alpha+1}{\sqrt{5}}$

$\therefore$ P lies on the line $2x - y + \alpha = 0$ at a distance $\frac{\alpha+1}{\sqrt{5}}$ units from S.

$\frac{x-0}{\frac{1}{\sqrt{5}}} = \frac{y-\alpha}{\frac{2}{\sqrt{5}}} = \pm\frac{(\alpha+1)}{\sqrt{5}}$

$(x, y) = \left(\frac{\alpha+1}{5}, \alpha + \frac{2(\alpha+1)}{5}\right)$ or $(x, y) = \left(-\frac{(\alpha+1)}{5}, \alpha - \frac{2(\alpha+1)}{5}\right)$

$x = \frac{\alpha+1}{5}, y = \frac{7\alpha+2}{5}$ or $x = \frac{-\alpha-1}{5}, y = \frac{3\alpha-2}{5}$

$\alpha = 5x - 1 = \dfrac{5y - 2}{7}$

$\alpha = -5x - 1 = \dfrac{5y + 2}{3}$

$35x - 7 = 5y - 2$ $\qquad\qquad$ $3x + y + 1 = 0$

$7x - y - 1 = 0$

PART – C

13. **2.00**

Foot of perpendiculars from any point on the circumcircle of the triangle to its sides are collinear. (simpson's line).

14. **9.00**

$CD^2 = CE \times CA \Rightarrow 36 = 3CA \Rightarrow CA = 12$

Also $BA = BD$,

Also $BC^2 = AB^2 + AC^2$

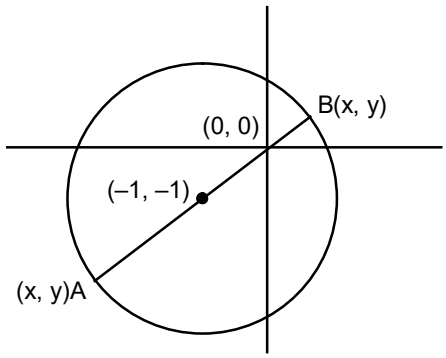
$\Rightarrow (x + 6)^2 + x^2 + 12^2 \Rightarrow 12x = 108 \Rightarrow x = 9.$

15. **6.00**

$a = \left(2 + \sqrt{2}\right)^2$ and $b = \left(2 - \sqrt{2}\right)^2$

$\therefore a + b = 12$

$\Rightarrow \dfrac{a + b}{2} = 6.$



16. **5.00**

For a given circle area of equilateral triangle would be maximum

$\Rightarrow R = \dfrac{abc}{4\Delta} = \dfrac{a^3}{4 \times \dfrac{\sqrt{3}}{4} a^2} = \dfrac{a}{\sqrt{3}}$

$\Rightarrow a = R\sqrt{3} = 2\sqrt{3}$

Hence $2s = 3a = 6\sqrt{3}$

Hence $[s] = 5$.

17. **1.25**

The equation of PQ is $4x + 4y = 7$.

18. **4.00**

Orthocentres of triangles formed by three tangents and corresponding normals to a parabola are equidistant from axis of parabola.